



Course Prepared by
IITian & Security Architect

PROGRAM DURATION

Course Duration	5 Months
Project Learning	3 Months
Internship	2 Months

ADVANCED CYBER SECURITY CERTIFICATION PROGRAM

Learn. Break. Defend. Report. Secure.

SKILL IT Education presents a structured, job-oriented career pathway in Cyber Security — from ethical-hacking fundamentals through penetration testing, cryptography and AI-powered threat defence, reinforced with hands-on labs and real attack simulations.

PROGRAM HIGHLIGHTS

- ✓ 6 Core Cyber Security Modules
- ✓ 5+ Hands-On Labs & AI-Assisted Capstone
- ✓ Penetration Testing, Cryptography & AI Defence
- ✓ Real-Time 2-Month Internship

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YOUR CYBER SECURITY JOURNEY



WHY THIS MATTERS

Cyberattacks are growing faster than the workforce trained to stop them. Organisations need professionals who can find vulnerabilities before attackers do — and use modern AI-driven tools to detect and respond to threats in real time.

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01 Program Overview

Build Your Cyber Security Career

What This Program Covers

This program takes you step-by-step from networking and ethical-hacking fundamentals to professional-grade penetration testing, cryptography and AI-powered threat defence — preparing you to secure real systems, not just study theory.

Why This Matters

Cyberattacks are growing faster than the workforce trained to stop them. Organisations need professionals who can find vulnerabilities before attackers do — and use modern AI-driven tools to detect and respond to threats in real time.

COURSE DURATION

5 Months

PROJECT LEARNING

3 Months

INTERNSHIP

2 Months

PROGRAM ROADMAP

1	Cybersecurity & Ethical Hacking Foundations	20 Hrs
2	Reconnaissance, Scanning & Enumeration	30 Hrs
3	System Hacking, Malware & Social Engineering	30 Hrs
4	Penetration Testing & Web Exploitation	60 Hrs
5	Advanced Exploitation, Cryptography & Mobile Security	20 Hrs
6	AI-Powered Cyber Defence & Security Operations	20 Hrs

02 Why Skill IT Education

Trusted by Thousands of Learners

15+

Years of Human Excellence

650+

Learners Trained

345+

Hiring Partners

180 Hrs

Hands-On Curriculum

Recognised & Industry-Aligned

Skill IT Education is recognised for excellence in professional education and training, with a curriculum aligned to NASSCOM industry standards to help close the national cybersecurity skills gap.

TOP REASONS TO CHOOSE US

Industry-Aligned Curriculum

Benchmarked against global cybersecurity accreditation bodies.

Mentored by Experts

Learn from certified security professionals & AI architects.

Real-Time Internship

Apply skills on live assignments during a 2-month internship.

Placement Support

A dedicated team guides you from resume to final interview.

KEY HIGHLIGHTS

- Flexible learning — repeat sessions, switch batches anytime
- Lifetime access to learning resources and lab environments
- Exclusive access to a Cyber Security virtual practice lab
- Minimum of 5 hands-on projects across every learner's portfolio
- Active learning community of mentors, learners and alumni
- Dedicated placement assistance from day one

03 Module 1 — Cybersecurity & Ethical Hacking Foundations

20 Hrs — LEARNING HOURS

WHAT YOU WILL LEARN

- Cybersecurity vs Information Security
- Types of Hackers & Threat Actor Profiles
- Cyber Kill Chain Concept & Attack Lifecycle
- Networking Fundamentals — OSI Model, Ports & Protocols
- 3-Way & 2-Way Handshake, Networking Devices
- Linux & Windows Command-Line Essentials
- VirtualBox & Kali Linux Lab Setup
- Vulnerable Server & Windows Installation for Practice
- Cyber Laws & Organisational Security Policies

TOOLS & PLATFORMS

Kali Linux

Parrot Security

VirtualBox

Linux/Windows CLI

✓ LEARNING OUTCOME

Students build a solid foundation in networking, operating systems and ethical-hacking methodology, and set up a fully functional penetration-testing lab.

04 Module 2 — Reconnaissance, Scanning & Enumeration

30 Hrs — LEARNING HOURS

WHAT YOU WILL LEARN

- Introduction to Reconnaissance & Intelligence Gathering
- Open-Source Intelligence (OSINT) Techniques
- WHOIS, Domain Registration & DNS Analysis
- Port Scanning & Advanced Scanning Techniques
- CVE, CVSS, CWE, NIST & NVD Vulnerability Databases
- Technology Stack Identification
- FTP, SMTP & SMB Enumeration
- Email Discovery & Verification Techniques
- Passive vs Active Reconnaissance
- Vulnerability Assessment Using Automated Tools

TOOLS & PLATFORMS

OpenVAS

Kali Linux

Parrot Security

✓ LEARNING OUTCOME

Students learn to profile targets, map attack surfaces and identify exploitable vulnerabilities using industry-standard scanning methodologies.

05 Module 3 — System Hacking, Malware & Social Engineering

30 Hrs — LEARNING HOURS

WHAT YOU WILL LEARN

- Windows Hacking & Privilege Escalation Techniques
- Malware Concepts & Types of Malware
- Windows Login Bypass Techniques
- Packet Analysis & Network Traffic Inspection
- Malware Creation Using Kali Tools (Controlled Lab Use)
- Anti-Malware Software & Defensive Techniques
- Server Hacking Fundamentals
- Social Engineering Concepts & Attack Phases
- Types of Social Engineering & Social Media Hacking

TOOLS & PLATFORMS

OpenVAS

Kali Linux

Parrot Security

✓ LEARNING OUTCOME

Students learn to compromise, analyse and defend systems against malware and social-engineering attacks, and understand the human layer of cybersecurity.

06 Module 4 — Penetration Testing & Web Exploitation

60 Hrs — LEARNING HOURS

WHAT YOU WILL LEARN

- DoS/DDoS Concepts & Attack Techniques
- Firewall Configuration & Wireless Encryption
- Session Hijacking Concepts & Cookie Stealing
- IDS, IPS, Firewall & Honeypot Evasion
- Web Server & Web Application Attack Methodology
- SQL Injection Concepts, Types & Injection Tools
- Cross-Site Scripting (XSS)
- Active Directory Concepts & Attack Techniques
- Wireless Network Security & WiFi Attacks
- Cloud Security Concepts & Cloud Service Models

TOOLS & PLATFORMS

Wireshark

Nikto

Invicti

Cain & Abel

✓ LEARNING OUTCOME

Students learn to plan and execute end-to-end penetration tests across networks, web applications, wireless networks and Active Directory environments.

07 Module 5 — Advanced Exploitation, Cryptography & Mobile Security

20 Hrs — LEARNING HOURS

WHAT YOU WILL LEARN

- IoT Concepts & IoT Attack Vectors
- Cryptography & Encryption Fundamentals
- Steganography & Data-Hiding Techniques (Image, Audio, Text)
- Mobile Security Fundamentals & Android Hacking
- Password Cracking & Recovery Techniques
- Network Discovery & Host Scanning

TOOLS & PLATFORMS

Angry IP Scanner

John the Ripper

Aircrack-ng

✓ LEARNING OUTCOME

Students extend their offensive skillset to IoT, mobile, wireless and cryptographic systems, identifying vulnerabilities across emerging attack surfaces.

08 Module 6 — AI-Powered Cyber Defence & Security Operations

20 Hrs — LEARNING HOURS

WHAT YOU WILL LEARN

- AI & Machine Learning Fundamentals for Security
- AI-Powered Threat & Anomaly Detection
- Phishing Detection & Email Security Using AI
- Malware Analysis with AI/ML-Based Techniques
- Automated Incident Response & SOC Workflows
- Vulnerability Classification & CVSS Scoring
- Adversarial Attacks & Securing AI Systems
- Professional Security Reporting & Executive Documentation

TOOLS & PLATFORMS

AI Threat-Detection Platforms

SOC Tooling

Reporting Frameworks

✓ LEARNING OUTCOME

Students learn to apply AI and machine learning to strengthen threat detection, automate incident response and defend AI systems themselves.

09 Cyber Security Technology Stack

The complete toolset used across every module of the program

Operating Systems & Lab

- Kali Linux
- Parrot Security
- VirtualBox
- Windows/Linux CLI

Vulnerability & Network Scanning

- OpenVAS
- Angry IP Scanner
- Wireshark
- Nikto

Exploitation & Password Tools

- Invicti
- Cain & Abel
- John the Ripper
- Aircrack-ng

Security Standards & Frameworks

- CVE
- CVSS
- CWE
- NIST/NVD

Cryptography & Steganography

- Encryption Toolkits
- Steganography Utilities
- Data-Hiding Tools

AI & Security Operations

- AI Threat-Detection
- SOC Automation
- Reporting Frameworks

10 Project-Based Learning

Every module is reinforced with hands-on work — a minimum of five projects across the program

01 Reconnaissance & Vulnerability Assessment Report

Perform OSINT, scanning and enumeration against a lab target, then document findings mapped to CVE/CVSS/CWE standards.

02 System & Malware Defence Exercise

Simulate system hacking and malware behaviour in an isolated lab, then build and test anti-malware defences.

03 Web Application & Network Penetration Test

Execute a full penetration test covering web applications, wireless networks and Active Directory.

04 AI-Assisted Threat Detection Mini-Project

Build a mini AI/ML-based workflow that flags phishing emails or anomalous network activity.

FINAL CAPSTONE

End-to-End AI-Assisted Security Assessment

Plan, execute and report a complete penetration test enhanced with AI-powered threat detection — covering reconnaissance, exploitation, proof-of-concept documentation, remediation guidance and retesting.

11 Learning Outcomes

What you will be able to do by the end of the program

- ✓ Understand networking, operating systems and the ethical-hacking methodology
- ✓ Identify and classify vulnerabilities using CVE, CVSS, CWE and NIST standards
- ✓ Recognise and defend against social-engineering attacks
- ✓ Exploit and secure wireless networks and Active Directory environments
- ✓ Apply cryptography, steganography and data-hiding techniques
- ✓ Use AI/ML techniques for threat detection, phishing detection & malware analysis
- ✓ Write professional, client-ready security assessment reports
- ✓ Perform reconnaissance, scanning and OSINT-based intelligence gathering
- ✓ Execute system hacking and analyse real-world malware behaviour
- ✓ Plan and run full-scope penetration tests on networks & web applications
- ✓ Assess cloud security posture and common cloud attack vectors
- ✓ Test mobile and IoT systems for security weaknesses
- ✓ Support automated incident response and SOC workflows
- ✓ Work confidently with industry-standard cybersecurity tools & platforms

12 Career Opportunities

Build your career in cyber security

Security Operations

- SOC Analyst
- Security Analyst
- Incident Response Analyst
- Threat Intelligence Analyst

Offensive Security

- Penetration Tester
- Ethical Hacker
- Vulnerability Assessor
- Red Team Associate

Network & Cloud Security

- Network Security Engineer
- Cloud Security Engineer
- Security Consultant
- Security Compliance Analyst

AI-Driven Security

- AI Security Analyst
- Threat Detection Engineer
- Security Automation Engineer
- SOC Automation Specialist

Career outcomes depend on individual skills, project execution, interview performance, experience level and market requirements.

13 Technical Assessment & Industry Certifications

Measure your skills. Prove your skills.

Weekly Assessments

- Regular technical assessments to measure understanding of recently completed topics.

Module-End Assessments

- Comprehensive technical evaluation after completion of each major module.

Practical / Lab Assessments

- Students demonstrate their ability to execute attacks and defences in real lab environments.

Project Assessments

- Evaluation of project methodology, execution, documentation and reporting quality.

CERTIFICATION PATHWAYS ALIGNED WITH COURSE CONTENT

CEH — Certified Ethical Hacker

CompTIA Security+

CompTIA PenTest+

OSCP

ISC2 Certified in Cybersecurity (CC)

AWS Certified Security – Specialty

Microsoft Security Operations Analyst

Certification Notice

The curriculum is aligned to help students prepare for these external certifications. Registration, fees and eligibility for any external exam are managed directly by the respective certifying body and may change over time. SKILL IT Education awards its own course-completion certificate upon successful completion of the program.

14 Internship, Industry Exposure & Placement Support

After successful course completion and project evaluations, eligible learners progress into a real-time, 2-month internship — gaining practical experience across Penetration Testing, SOC Operations, Cloud Security and AI-Assisted Threat Detection.

01 Career Guidance

Help identifying suitable security job roles and career paths.

02 Resume Support

Personalised resume reviews and improvement tips.

03 Project Mentorship

Live guidance on industry-relevant security projects.

04 Interview Preparation

Practice sessions with common technical & HR questions.

05 Doubt Resolution Sessions

Regular Q&A sessions with mentors and security experts.

06 Alumni Network

Connect with placed alumni and learn from their journey.

Complete Career Transition Support

Skill IT Education's Placement Assistance Team works closely with every learner — offering personalised career guidance, mock interviews and placement support through a strong network of 1000+ hiring partners across leading IT, product and consulting companies.



WHY CHOOSE SKILL IT EDUCATION

DON'T JUST LEARN CYBER SECURITY. LEARN TO ATTACK, DEFEND & SECURE.

Learn. Break. Defend. Report. Secure.

SKILL IT Education builds cybersecurity professionals, not just tool users. Through an industry-driven curriculum, hands-on lab work and a real-time internship, every learner graduates with a portfolio of documented security assessments, verified skills and the confidence to walk into technical interviews and deliver from day one.

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Let's build a safer future.

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